IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Kingdom of Bahrain
- Organization(s) or entity(s) responsible for the preparation of the report*:

  Name: The UNESCO Regional Centre for Information and Communication Technology (RCICT)

  Website: https://regcict.com/

  Email address*:

  Phone number*:

  Please describe the role/mandate of your organization: focal point

- Officially designated contact person(s) who completed this survey:

  Full name*: Dr. May Ahmed Shamandy

  Position*: Acting Director

  Email address*:

- Full Name(s) of designated official(s) certifying the report:
- Brief description of the consultation process established for the preparation of the report:
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:

Name of the organisation	Website	Email	Sector
Nasser Centre for Science and Technology	www.nvtc.edu.bh		Private
Higher Education Council	https://hec.gov.bh		Public
Bahrain Centre for Strategic, International and Energy Studies Derasat	www.derasat.org.bh		Other
Information & eGovernment Authority (iGA)	https://www.iga.gov.bh/en/		Public
UNESCO Regional Centre for Information and Communication Technology (RCICT)	https://regcict.com/		Public

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The ministries and national authorities/entities involved in promoting the 2021 Recommendation include the Ministry of Education, the Higher Education Council, Nasser Centre for Science and Technology, the Labour Fund (Tamkeen), and the Ministry of Labour. Additionally, higher education institutions, the Bahrain Centre for Strategic, International, and Energy Studies (Derasat), the Information & eGovernment Authority (iGA), and the UNESCO Regional Centre for Information and Communication Technology (RCICT) have also contributed to this effort.

Nasser Centre for Science and Technology (NCST)	Yes, the ministries and the national authorities/entities involved in the promotion of the 2021 Recommendation are : Ministry of Education, Higher Education Council, Nasser Centre for Science and Technology (NCST), the Labour Fund (Tamkeen) and the Ministry of Labour.
Higher Education Council	YES. Higher education Institutions
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	N/A
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Yes. Ministry of Education, Nasser Centre for Science and Technology(NCST), Higher Education Council, Bahrain Centre for Strategic, International and Energy Studies (Derasat), The Information & eGovernment Authority (iGA)

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)?

Arabic and English

Nasser Centre for Science and Technology	Yes, It was provided in Arabic which is the national language of Bahrain, along with English and many other languages.
Higher Education Council	Yes. Arabic
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES. Arabic.
Information & eGovernment Authority (iGA)	--
The UNESCO Regional Centre for Information and Communication Technology (RCICT)	YES. Arabic and available in English too.

1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

The Nasser Centre for Science and Technology(NCST) organizes and foresees several activities that contribute to the promotion of open science. One of the key initiatives is providing education and science for all, with a focus on offering advanced technical and vocational education in line with the Fourth Industrial Revolution. The center organises annual summer workshops targeting all segments of society to raise awareness about the importance of technical and industrial education. These workshops cover a wide range of topics, such as mechanical and electrical maintenance, cybersecurity, artificial intelligence tools, and soft skills development.

Over the past years, the center has organised numerous lectures, workshops, and field visits, benefiting over 2,000 participants. The center also facilitates field visits for students from local and international institutions, providing them with hands-on experiences and fostering collaboration in the field of open science. Furthermore, the Nasser Centre conducts annual virtual orientation meetings for citizens, students, and parents, guiding them on how to access and benefit from its services.

Additionally, the center runs a Student Clubs Programme that focuses on developing students' skills and talents in cultural, social, and sports activities while enhancing their leadership and communication skills. It also organizes scientific and professional exhibitions that allow students to interact with innovations and technologies, reinforcing the concept of open science. To encourage practical application of knowledge, the center involves students in external scientific competitions that foster critical thinking, problem-solving, and innovation.

The Nasser Centre also organizes educational field visits for students to gain practical experience and interact with experts in various fields. To support open education, the center has created a student portal that offers easy access to educational resources, promoting self-directed learning and contributing to the development of a sustainable scientific community.

The Higher Education Council **(HEC)** has also been active in promoting open science, such as inviting higher education institutions to participate in the eighth specialized session on "**Open Science: A Gateway to Accelerating Progress on the 2030 Agenda in the Post-Covid-19 Era**" in March 2021. Additionally, UNESCO's questionnaire on "Global Call for Best Practices in Open Science" was completed in May 2022.

The Information & eGovernment Authority (iGA) has supported open government data through awareness sessions, and the UNESCO Regional Centre for Information and Communication Technology (RCICT) has raised awareness about open educational resources (OER) and open science through activities with various entities in the Ministry of Education, the Higher Education Council, and private educational institutions in the Kingdom of Bahrain.

Nasser Centre for Science and Technology (NCST).	Nasser Centre for Science and Technology contributes to supporting the goals of open science through several aspects, including: 1. Providing Education and Science for All: The center's policy believes in the necessity of providing advanced technical and vocational education to all students in the Kingdom, in line with the developments of the Fourth Industrial Revolution, through transparent and fair admission policies. 1.01 Lectures, Workshops, and Scientific Field Visits Provided by Nasser Centre for Science and Technology for the General Public: Nasser Science and Technology Center organizes annual summer workshops to spread science and knowledge, targeting all segments of society to raise awareness of the importance of technical and industrial education. These workshops contribute to expanding the participants' knowledge and enhancing their skills by interacting with subject matter experts in various fields. The following is a summary and outline of some of the workshops, lectures, and field visits offered by the Nasser Center over the past years: - Workshops for the General Public in Various Specializations, Topics, and Skills:These include mechanical and electrical maintenance, cybersecurity, the use of artificial intelligence tools in various fields, moreover soft skills development that includes public

	speaking and so on. These online workshops, initiated by the Nasser Center four years ago, have benefited more than 2000 participants in their fourth season. This includes the lectures delivered by the Nasser AI Research and Development Center Public and Educational Contributions to emphasize the importance of artificial intelligence (AI) and related sciences provided to various entities like: West Riffa Girls School, French International School of Bahrain, Judging IoT-AI competition at University of Bahrain for 3 years, Judging Ministry of Education Secondary Girls schools' qualification, Sheikha Fadia al Saad al Sabah Scientific Research and Projects Competition, Lecture at AlAhlia University , Lecture at University of Bahrain Vocational studies, Lecture at University of Bahrain Statistics, Lecture at University of Bahrain Computer studies, Lecture at University of Technology Bahrain, Two Lectures at A.Rahman Kanoo Culture Centre, Lecture at Alumni Club Bahrain, Lecture for Shura Council members, Lecture for Shura Council General Assembly, Lecture for House of Representatives members, Lecture for Alba employees, Lecture at Education Excellence, Judging at Bahrain Association of Bankers, Lecture at NHRA conference, Lecture at Bahrain the Arab International Cybersecurity Conference, Lecture at BITEX conference, West Riffa Girls School, Lecture for Ministry of Finance employees, Lecture for Bahrain Customs employees and Lecture for Royal Medical Services employees. 1.02 Field visits organized by Nasser Centre for Scientific and Technological for university and school students, and various institutions in the Kingdom of Bahrain or from abroad: The organization of field visits by Nasser Center for school and university students, whether from within or outside the Kingdom of Bahrain, plays a central role in achieving the goals of the Open Science Programme through collaboration. It provides students with a unique and direct educational experience that allows them to interact with real scientific and professional environments, enhancing their understanding and enriching their educational experience. These visits also contribute to spreading the culture of open science by encouraging the exchange of ideas and knowledge among students from diverse educational and cultural backgrounds, thereby reinforcing the values of scientific collaboration and promoting the development of a conscious generation eager to advance society through open science and openness to new scientific fields. 1.03. Organizing annual virtual orientation meetings for all citizens in coordination with the Ministry of

	Education, to introduce the Centre, its curricula, and its educational services for future interested students who wish to enroll in the center. 1.04. Organizing annual virtual orientation meetings for all students and parents to educate parents and guide them on how to access and benefit from the services provided, ensuring that the goals of these services are achieved and that they reach all individuals, including students and parents. This also ensures the continuous communication and awareness of the importance of these services, as well as informing parents and students about their rights and responsibilities concerning the data and systems used to deliver various services. 2. Nasser Centre for Science and Technology Student Clubs Programme, which focuses on developing students' skills, honing their talents, and maximizing their creative abilities. The center has been offering student club activities for over 6 years to nurture and encourage talents and hobbies in various cultural, social, and sports areas, while also enhancing their social skills which includes leadership, communication skills and other competences among students, who have the ability to select their club of interest, thus allowing freedom of choice. 3. The Nasser Centre attaches utmost importance to organizing scientific and professional exhibitions for its students, as they play a crucial role in achieving the goals of the Open Science Programme. These exhibitions provide students with a platform to interact directly with the latest innovations and technologies in various fields. Through these exhibitions, students have the opportunity to explore real scientific applications and professional skills, expanding their horizons and increasing their scientific curiosity. These exhibitions also reinforce the concept of open science, encouraging the exchange of knowledge and facilitating access to information, enabling students to understand the importance of scientific collaboration and experience sharing to build a strong knowledge-based society in Bahrain. 4. Involving Students in External Competitions to Equip Them with Knowledge and Apply the Theoretical Knowledge Learned at the Nasser Centre Practically in Finding Innovative Solutions The Nasser Centre believes that involving its students in scientific competitions is a fundamental pillar in motivating them to apply scientific knowledge to practical challenges, enriching their experiences and encouraging innovation and creativity. These competitions provide students with an opportunity to expand their scientific horizons through competition in environments that enhance their critical thinking and problem-solving abilities.

	Furthermore, these competitions promote the principle of open science by encouraging students to share their ideas and research findings with a broader community, contributing to the dissemination of knowledge and facilitating access to it. They support the creation of an open, collaborative scientific culture in the Kingdom of Bahrain, based on continuous development and the exchange of scientific experiences. Educational Field Visits Provided by the Centre for Students Inside and Outside Kingdom of Bahrain The Nasser Centre believes that the field visits it offers to its students provide an opportunity to gain direct practical experience and learn about the latest engineering technologies and real-world applications of academic concepts. Through these visits, students interact with professionals and experts in their fields, expanding their knowledge and opening new horizons for developing their skills and enhancing their ability to innovate. 5. Student Portal: An electronic portal was created to support open education, featuring a comprehensive educational library and a platform for managing educational content, allowing students easy access to academic resources anytime and anywhere. This portal enables students to engage in self-directed learning and interact with educational content according to their needs and interests. It also promotes the principle of open science by providing resources to all without restrictions, contributing to the creation of a diverse and sustainable scientific community.
Higher Education Council	• In March 2021, higher education institutions were invited to participate in the eighth specialized session titled : 'Open Science: A Gateway to Accelerating Progress on the 2030 Agenda in the Post-Covid-19 Era'. This event was organized by the UNESCO Regional Office for Science in the Arab States - Cairo, in collaboration with the United Nations Economic and Social Commission for Western

	Asia (ESCWA) and the United Nations Environment Programme (UNEP) as part of the Arab Forum for Sustainable Development 2021. • In May 2022, UNESCO's questionnaire entitled (UNESCO's Global Call for Best Practices in Open Science) was completed.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	No
Information & eGovernment Authority (iGA)	The Information & eGovernment Authority (iGA) has promoted open government data through awareness sessions, which might be considered as part of open science.
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Awareness raising activities on OER and Open Science took place with many entities in the Ministry of Education including all K12 teachers and students, Higher Educational Council, and private educational institutions in the Kingdom of Bahrain.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[ii], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

The Nasser Centre for Science and Technology (NCST) is actively working to incorporate the values and principles of open science through various initiatives. One of its key actions is the development of a comprehensive student portal that will include open data to support research and innovation, providing scientific information and data for students, alumni, and researchers to utilize in their projects and studies.

Future plans include expanding this portal to enable education for all, regardless of age or academic background, and improving access to educational resources. These efforts aim to promote open science by supporting continuous learning, disseminating knowledge across society, and fostering innovation to contribute to sustainable development.

In contrast, the Higher Education Council (HEC) has not yet outlined specific actions but acknowledges that some principles of open science, such as promoting multilingualism in scientific publications, fostering bibliodiversity, and respecting the rights and needs of communities, are already embedded in current practices at higher education institutions. These values will be further examined and developed as part of ongoing efforts to strengthen the integration of open science practices in publicly-funded research.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) has not yet implemented explicit actions toward open science, largely due to a lack of awareness and the nascent stage of the research and scientific culture in the country. However, research institutions and universities have taken significant steps to indirectly incorporate open science principles. The Information & eGovernment Authority (iGA) has not specified any actions in this area.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) has been active in promoting open science values by delivering workshops and training on Open Educational Resources (OER) to all public schools in Bahrain through the Ministry of Education's related departments. Plans for 2025 include delivering awareness workshops on the definition, values, and applications of open science, as well as holding OER training workshops for the Higher Education Council.

	Nasser Centre for Science and Technology(NCST) aims to develop the student portal into a comprehensive platform that includes open data supporting research and innovation, providing scientific information and data that students, alumni and researchers can use in their projects and studies. Future plans also include creating a comprehensive portal that enables education for all, regardless of age or academic background, improving direct and easy access to educational resources. These projects enhance the concept of open science by supporting continuous learning and disseminating knowledge across all segments of society, helping to build a knowledgeable, developed community capable of innovation and contributing to sustainable development.
Higher Education Council (HEC)	No. Some values and principles of open science are already embedded in the current practices of higher education institutions. These include promoting multilingualism in scientific publications and academic communications, fostering biodiversity through various publication formats and channels, and ensuring that the rights and needs of communities are respected. Moving forward, these values and principles will be further examined and developed as part of ongoing efforts to strengthen the integration of open science practices in publicly funded research.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO, there are no explicit action plans in place towards open science. This can mainly be attributed toward awareness, and the nascent stage of the research and scientific culture, at the country-wide level. However, research institutions and universities have taken significant steps in indirectly incorporating open science principles and provisions.
Information & eGovernment Authority (iGA)	--

UNESCO Regional Centre for Information and Communication Technology (RCICT)	Workshops and trainings on OER were delivered to all public schools in the Kingdom of Bahrain by Ministry of Education's related departments. There are also plans to deliver awareness workshops about Open Science definition, values and applications during 2025. Also, OER training workshops will be held for the Higher Educational Council.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy[iii], strategy or plan of action on science, technology and innovation (STI)?

YES / NO

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*
- does it include any of the following open science elements:
  - open access to scientific knowledge
  - open science infrastructure
  - open engagement of societal actors
  - open dialogue with other knowledge systems

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The Nasser Centre for Science and Technology has drawn up a national strategy on science, technology, and innovation (STI) under the theme "National Strategy for Science, Technology, and Innovation." The blueprint was adopted in 2020 under the authority of the Ministry of Education. The strategy's key areas focus on supporting open access and public engagement in research.

In addition, the Higher Education Council (HEC) has also developed the "National Strategy for Higher Education," which covers vision, key objectives, national aspirations for higher education in the Kingdom of Bahrain, challenges, and other related themes. This policy, updated for the period from 2014 to 2024, outlines Bahrain's educational and higher education framework.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) notes that while there is no separate policy dedicated to open science elements, these are incorporated under Bahrain's "Economic Vision 2030." Various government bodies have taken initiatives related to advancements in emerging technologies such as Blockchain, Artificial Intelligence (AI), Data Analytics, Robotics, IoT, and Smart Cities. However, not all aspects of open science have been fully addressed.

The Information & eGovernment Authority (iGA) runs the Technical Development Programme in collaboration with the Ministry of Labour. This initiative aims to enhance the skills of young Bahraini university graduates and job-seekers aged 21 to 30 in the Information Technology (IT) sector. Participants receive training, a salary, and internationally accredited professional certificates, enhancing their employment prospects in both the public and private sectors.

Furthermore, the UNESCO Regional Centre for Information and Communication Technology (RCICT) aligns with the government's approach to supporting open access and public engagement. RCICT hosted a regional workshop in collaboration with UNESCO in 2014, leading to the creation of an Open Educational Resources (OER) policy by the Ministry of Education.

The Centre also participated in the 2nd Arab Open Access Forum in 2021, where it held a workshop on OER and Creative Commons, benefitting approximately 260 participants from various Arab countries. Additionally, RCICT coordinated with the Arab League Educational, Cultural, and Scientific Organization (ALECSO) to host a virtual regional meeting in 2021, promoting the dissemination of open educational resources across Arab countries.

Nasser Centre for Science and Technology (NCST)	Title of the Policy: National Strategy for Science, Technology, and Innovation Authority in Charge: Ministry of Education Year of Adoption: 2020 Key Areas: Supports open access and public engagement in research Government Plan
Higher Education Council (HEC)	Policy title: National Strategy for Higher Education The authority in charge of development of the policy: Higher Education Council Adoption year and latest update: 2014 - 2024 Key areas/sections: Vision, Key Objectives, National Aspirations for Higher Education, Context and Challenges, Themes and Objectives. NATIONAL+STRATEGY+FOR+THE+HIGHER+EDUCATION-AR.pdf bahrain+higher+education+strategy+-+summary.pdf it includes any of the following open science elements: • open access to scientific knowledge • open science infrastructure
Bahrain Centre for Strategic, International and Energy Studies	YES. Although not separately, these policies are outlined under the “Bahrain Economic Vision 2030”. Different government bodies have taken initiatives towards making advancements in Blockchain, AI, Data Analytics, Robotics, IOT and Smart Cities. Link: Approach to New Emerging Techologies

(Derasat)	However, Not all of the elements of open science have been addressed adequately.
Information & eGovernment Authority (iGA)	The Technical Development Programme is conducted by the Information & eGovernment Authority, in cooperation with the Ministry of Labour to support new graduates and job seekers in the Information Technology (IT) field. The (IT) scheme is one of the strategic projects that aim to refine the skills of young Bahraini universities graduates and job seekers with technical specializations from those between the ages of 21 to 30 years who are registered in the Unemployment Insurance System to be human capital and distinguished and experienced national cadres, in line with the requirements of the labour market, and become the best choice in employment. The programme's participants are given an opportunity to obtain intensive training in the iGA's technical departments supported by a monthly salary, in addition to receiving internationally accredited professional certificates, raising their chances to be employed by either the public or private sector at the end of the programme. Link for register for this programme: https://www.iga.gov.bh/en/article/the-technical-development-program o open access to scientific knowledge (Yes)
UNESCO Regional Centre for Information and Communication Technology (RCICT)	- The Centre works in parallel with the Government Plan in which it supports open access and public engagement. - A regional workshop was held in cooperation with UNESCO under the patronage of Minister of Education on 22-24 September 2014, led by Mr. Miao Head of Technology and AI in Eucation UNESCO – Paris. - As a result of the regional workshop, an OER policy was written by designated authorities in the Ministry of Education in 2014. - RCICT participated in the 2nd Arab Open Access Forum in 2021 with a workshop about OER and Creative Commons for around 260 beneficiaries from different Arab countries. - In implementation of the recommendations of the 11th Arab Ministers of Education Conference, which was held in the Kingdom of Bahrain on November 6-7, 2019, on "Working to disseminate open educational resources in accordance with the standards adopted in Arab countries", and based on the vision and mission of the Arab League Educational, Cultural and Scientific Organization (ALECSO) in promoting quality and excellence in the field of disseminating the use of open educational resources, the Regional Center, in coordination with (ALECSO), took the initiative by

	holding a virtual regional meeting on February 23, 2021, on "Disseminating the Use of Open Educational Resources in Arab Countries".

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[iv] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES / PARTLY /NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

In the Kingdom of Bahrain, there are some policies and frameworks that address Open Science, although they are not fully aligned with the 2021 Recommendation on Open Science in all aspects. The Higher Education Council's policy, which is partly relevant, is based on the Constitution of the Kingdom of Bahrain, adopted on February 14, 2002. The Constitution addresses basic components of society in Articles 7 and 23, which provide a foundation for policies related to science, technology, and education. However, open science elements are not explicitly covered under this document.

After the recent independence of the Higher Education Council from the Ministry of Education, both administratively and financially, the Council is working on developing a modern legislative framework to govern higher education and scientific research, which will likely incorporate aspects of open science.

The Bahrain Centre for Strategic, International and Energy Studies (Derasat) reports that there is no national policy dedicated specifically to open science. However, related provisions can be found in the "Open Government Data Policy," the "Bahrain Open Data Portal," and data protection measures. While these policies address some aspects of open data, there has not been a centralised approach to open science at the national level, though plans to address this may emerge under Bahrain's Economic Vision 2030.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) reports that a committee for Open Educational Resources (OER) has been established within the Ministry of Education. Additionally, the concepts of OER and Open Science are integrated into the Ministry's strategies and planning, further contributing to the development of an Open Science framework in the country. However, more comprehensive national policies on Open Science are still under development.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council	Partly - Title of the policy or legal instrument: Constitution of the Kingdom of Bahrain. - Authority responsible for policy development : Government of the Kingdom of Bahrain. - Entities involved and development process : Ministries, government entities and the private sector - Year of adoption, 14-February-2002 - Year of last update: - - Main areas/sections of the policy, and elements of open science addressed: Basic components of society Article (7), and Article (23). - Link:*https://www.lloc.gov.bh/PDF/Constitution.pdf* Explanation of the reason and difficulties or obstacles in this area : Following the recent independence of the Higher Education Council from the Ministry of Education, both financially and administratively, and the cancellation of the previous regulations, the Council is in the process of developing a modern legislative framework to govern higher education and scientific research at higher education institutions. This new system aims to align with current developments and simultaneously address aspects related to open science.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. The only provisions in this regard are the “Open Government Data Policy”, “Bahrain Open Data Portal” and data protection measures. There has not been a centralized approach towards open science at the country level, and there are plans to address this under the economic visions
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	A committee for OER has been established in the Ministry of Education as well as the concepts of OER and Open Science are incorporated in the Ministry’s strategies and planning.

2.3 In your country, are there specific policy instruments[v] that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

There are limited specific policy instruments aimed at promoting Open Science. The Higher Education Council is in the process of developing regulations and guidelines for the operation of higher education institutions. While there is no current regulation explicitly focused on open science, the Council intends to consider such a regulation as it moves forward with its development of institutional frameworks.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) notes that there is no dedicated policy for open science. This is due to a lack of awareness about open science principles, a nascent research ecosystem, and a greater focus on economic growth and digital transformation, which currently take precedence over academic and research frameworks related to open science.

However, the UNESCO Regional Centre for Information and Communication Technology (RCICT) highlights an important development. The Ministry of Education in Bahrain published an Open Educational Resources (OER) policy in 2015, which is reviewed and updated annually.

This policy promotes the licensing of educational, training, and research content produced under the Ministry's umbrella with open licences, specifically Creative Commons licences. It encourages the free distribution, use, modification, and sharing of educational content, thereby supporting open access to knowledge and contributing to the broader dissemination of education and training. This policy represents an important step in promoting Open Science, although it is focused specifically on educational materials.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No Given that the Higher Education Council is in the process of developing regulations and guidelines to organize the operations of higher education institutions, a regulation to promote open science within these institutions will be considered.

Bahrain Centre for Strategic, International and Energy Studies Derasat	NO. Lack of awareness of the open science principles, nascent research ecosystem, and increased focus on economic growth and digital transformation supersede academic and research frameworks aims. This appears to be a secondary concern, and is subsequent.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	The Ministry of Education in the Kingdom of Bahrain published in 2015 an OER policy for teachers & students. It gets reviewed and updated annually or whenever needed. This policy aims to license all educational, training, and research contents produced under the Ministry of Education's umbrella with open licences (Creative Commons). The Ministry of Education encourages the free distribution, use, modification, and sharing of these educational contents, in accordance with the conditions specified in the open licence. The policy promotes open access to knowledge and contributes to the widespread dissemination of education and training.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

YES / UNDER DEVELOPMENT /NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

There is currently no national implementation plan, strategy, policy, or roadmap specifically for the national implementation of Open Science in alignment with the 2021 Recommendation. The Higher Education Council is in the process of developing a national strategy for Higher Education and Scientific Research for the period 2025-2035, which is expected to incorporate the implementation of open science. However, some higher education institutions have already adopted practices related to open science.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) points out that there is a lack of awareness about open science principles, a nascent research ecosystem, and a greater emphasis on economic growth and digital transformation over academic and research frameworks. Consequently, open science remains a secondary concern.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) also reports that there is no formal national plan for implementing Open Science at the country level. However, RCICT supports the Ministry of Education in advancing Bahrain's educational system and improving student learning outcomes, teacher performance, and community awareness of educational resource policies. These efforts indirectly contribute to the goals of Open Science by promoting digital empowerment and open access to educational resources.

Nasser Centre for Science and Technology (NCST)	Both the Ministry of Labour and the Labour Fund (Tamkeen) have drawn up a plan and strategy to develop the skills and knowledge of individuals allowing them to gain certifications and technical skills enabling them to explore opportunities in their careers. The aim of these strategies, policies and plan is to create additional job opportunities for citizens.
Higher Education Council (HEC)	**Under development** There is currently no national plan, strategy, or policy in place to implement open science at the national level; however, some higher education institutions have adopted practices related to open science. Additionally, the Higher Education Council is in the process of developing a national strategy for Higher Education and Scientific Research for the period 2025-2035, which will incorporate the implementation of open science.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. Lack of awareness of the open science principles, nascent research ecosystem, and increased focus on economic growth and digital transformation over academic and research frameworks. This appears to be a secondary concern, and is subsequent.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Under development There is currently no national plan, strategy, or policy in place to implement open science at the national level; however, some practices related to open science. RCICT supports the Ministry of Education in the Kingdom of Bahrain to achieve its vision of developing a good educational system to reach a high level of excellence and creativity.

	To achieve this vision, the Government has advanced its educational system and improved the quality of students' learning outcomes and empowered them digitally, developed the level of teachers' performance in teaching and learning, and increased community awareness of educational resources policies.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science**Error! Bookmark not defined.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

Several institutions and organizations have initiatives that promote Open Science in line with the 2021 Recommendation at the institutional level. For example, Derasat, the Bahrain Centre for Strategic, International, and Energy Studies, has established collaborative partnerships with academic and research units worldwide, which enhances the co-production of knowledge and fosters the exchange of ideas and expertise. The think tank's E-Library also contributes to public accessibility to its research.

The University of Bahrain supports open science by promoting open access through its dedicated journals and a robust scientific research department, while also funding research and academic dialogue as part of its key activities.

Additionally, the UNESCO Regional Centre for Information and Communication Technology (RCICT) plans to deliver workshops to raise awareness about the definition, values, and applications of Open Science, and is currently collaborating with Category II UNESCO centers to further these goals.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	N/A
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES, at the institution and university level, certain measures are in place that partially foster open science, for instance: Institution: Derasat Has collaborative partnerships with multiple academic and research units worldwide, enhancing co-production of

	knowledge and exchange of ideas and expertise. This think tank's E-Library improves public accessibility to its research. University of Bahrain Promotes open access through its own dedicated journals, a robust scientific research department, and funding research and dialogue in academia as part of its key activities.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Plans to deliver awareness workshops about Open Science definition, values and applications. Collaboration with Category II UNESCO centres is in progress.

2.6 In your country, are there specific funding mechanisms[vi] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

In the Kingdom of Bahrain, there are several scholarship opportunities for students, both nationally and internationally, that indirectly support Open Science by fostering academic excellence. These scholarships are provided by various entities, such as the Ministry of Education, the Al-Mabbarah Alkhalifia Foundation, and the Crown Prince International Scholarship Programme (CPISP), among others, and are targeted towards undergraduate and postgraduate studies.

Additionally, public schools in Kingdom of Bahrain are fully funded through the budget allocated by the Ministry of Finance and National Economy, post-approval. However, at the national level, there are no specific funding mechanisms dedicated solely to open science.

While private higher education institutions receive funding for scientific research through a requirement to allocate at least 3% of their annual net revenue, as per Decision No. 206 of 2023 by the Higher Education Council (HEC), these funds are not specifically designated for Open Science.

Similarly, institutions like Bahrain Centre for Strategic, International and Energy Studies (Derasat) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) do not have specific funding for Open Science initiatives.

Nasser Centre for Science and Technology (NCST)	Many scholarship opportunities exist for excellent students both notionally and internationally. These scolarships are provided by various entities, such as the Ministry of Education, Al-Mabbarah Al khalifia foundation, the Crown Prince International Scholarship Programme (CPISP), which are all supported by various Governments. ??? These scholarships are targeted towards undergraduate and postgraduate studies to explore further knowledge. Public schools are fully covered by the budget provided by The ministry of Finance and National Economy. .
Higher Education Council (HEC)	No There are no specific funding mechanisms or allocations for open science yet. However, certain activities related to open science in private higher education institutions are financed through funds designated for scientific research. Edict No. 206 of 2023 issued by the Higher Education Council (HEC), makes it mandatory for private higher education institutions to allocate a minimum of 3% of their annual net revenue for scientific research funding in general.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. There is no national dialogue on the open science aspects.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	No

2.7 In your country, have open science practices[vii] been integrated in existing research funding mechanisms?

YES /NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Open Science practices have not been integrated into existing research funding mechanisms. The Higher Education Council does not include specific open science practices in its funding framework. However, some higher education institutions, such as Kingdom University, have adopted activities related to open science, like supporting the publication of peer-reviewed journals and ensuring their public accessibility.

Despite these efforts, there is no formal integration of open science practices into the national research funding structure. Institutions such as Bahrain Centre for Strategic, International and Energy Studies (Derasat) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) also do not have Open Science practices embedded in their funding mechanisms, largely due to limited awareness and understanding of Open Science.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No There are no specific practices integrated into the existing research funding mechanisms. However, some higher education institutions have adopted certain activities related to open science. For instance, some aspects of scientific research funding can be aligned with open science, such as supporting the publication of peer-reviewed journals and ensuring their accessibility to the public, like the journals published by Kingdom University **Research - Kingdom University**
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. This is mainly due to limited awareness and understanding of open science
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	No

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? *(ref.: (ii) 17.i)*

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Efforts to foster equitable public-private partnerships on Open Science in the Kingdom of Bahrain are underway and planned for the near future. The Higher Education Council has outlined an initiative in its Operational Plan for 2023-2026 to create a National Research Network, aiming to promote collaboration among researchers, academic institutions, and industry stakeholders to drive innovation and enhance the impact of scientific research.

Public-private partnerships are also a key strategy within Bahrain's Economic Vision 2030, and while there is no explicit open science mandate, policies are in place to encourage these partnerships for both economic and academic purposes.

Additionally, the National Research Strategy 2014-2024 further supports these collaborations. The UNESCO Regional Centre for Information and Communication Technology (RCICT) is also actively engaging in communication with UNESCO's CI office on the implementation of Open Educational Resources (OER) and Open Science in the fields of ICT and education.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) is collaborating with entities that share similar developmental visions and goals to build research networks in educational technology and creativity, benefiting from the experiences of universities and research institutions through the Ministry of Education and Higher Education Council. These initiatives reflect a growing commitment to fostering equitable public-private partnerships in Open Science by the end of 2025.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	Yes One of the initiatives outlined in the Higher Education Council's Operational Plan for 2023-2026 is the creation of a National Research Network. The aim is to promote collaboration among researchers, academic institutions, and industry stakeholders, thereby driving innovation and strengthening the impact of scientific research in the Kingdom of Bahrain.
Bahrain Centre for Strategic, International and Energy Studies	YES, Public-Private partnerships are a key strategy of the Bahrain Economic Vision 2030. Although an open science mandate is not explicity present, policies are in place to promote these partnerships for all purposes albeit economic

(Derasat)	activity or academic research. Furthermore, they are also part of the National Research Strategy 2014-2024
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Yes. Communication is always ongoing with UNESCO CI office on OER and Open Science (OS) implementation in ICT and Education. Also, RCICT is working with entities that share similar developmental visions and goals, seeking to build research networks in the fields of educational technology and creativity, in addition to benefitting from experiences of various universities and research institutions through the Ministry of Education and the Higher Educational Council.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy *(ref.: (ii) 17.j, 23.c)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

Currently, there is no national monitoring framework for Open Science in the Kingdom of Bahrain, nor are there specific indicators related to Open Science included in the national monitoring and evaluation framework for Science, Technology, and Innovation (STI) policy. The Higher Education Council has indicated that this may be considered in the future, but at present, no formal monitoring system exists. Similarly, institutions such as Bahrain Centre for Strategic, International and Energy Studies (Derasat) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) do not have specific frameworks or indicators for Open Science.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No This will be taken into consideration in the future.
Bahrain Centre for Strategic, International and Energy Studies	No

(Derasat)	
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	N/A

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.a)*

The latest official data on Bahrain's research and development (R&D) expenditure as a percentage of its national Gross Domestic Product (GDP) is 0.1%, based on 2014 statistics. This data is sourced from the World Bank and the UIS (UNESCO Institute for Statistics) and reflects the R&D expenditure relative to the country's GDP for that year.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	R&D expenditure as a percentage of GDP 0.10116%-2014 UIS Statistics
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	As of 2014, the R & D expenditure as a % of GDP stands at 0.1%. Source: World Bank
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.b)*

The latest official data on internet access in the Kingdom of Bahrain indicates that 99% of the population had access to the internet as of the beginning of 2024, according to the "Digital 2024: Bahrain" report by DataReportal.
Additionally, other sources show that as of 2023, the figure for internet access stood at 100%, according to the World Bank. The Telecommunications Regulatory Authority (TRA) also reports that, as of 2022, 100% of the population had access to reliable internet and bandwidth. These statistics reflect Bahrain's high level of connectivity in recent years.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	Percentage of the population of the Kingdom of Bahrain who have access to the Internet 99.0% - of the total population as of the beginning of 2024: Digital 2024: Bahrain — DataReportal – Global Digital Insights
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	As of 2023, the figure was 100%, Source: World Bank
Information & eGovernment Authority (iGA)	The percentage of Kingdom of Bahrain is 100% for 2022. The source of the data is Telecommunications Regulatory Authority (TRA). Source : https://unstats.un.org/sdgs/dataportal/database
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

3.3 Are there national research and education networks (NRENs)[viii] active in the country? *(ref.: (iii) 18.c)*

YES / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

Currently, there are no national research and education networks (NRENs) fully operational in the Kingdom of Bahrain, though efforts are underway to establish one. The Higher Education Council has outlined the creation of a national research network as part of its operational plan for 2023-2026.
On the other hand, Bahrain Centre for Strategic, International and Energy Studies (Derasat) confirms that there are existing networks primarily accessible to educational campuses across the country. These networks are part of the design and implementation of a novel networking architecture for academic institutions in Bahrain, under the NREN section. While progress is being made, the full establishment of a national research network is still in development.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No Establishing a national research network is one of the projects being implemented as part of the initiatives and projects outlined in the Higher Education Council's operational plan for 2023-2026.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES. These network are mainly accessible to educational campuses across the country. Design and Implementation of NOVEL Networking Architecture for Academic Institutions in the Kingdom of Bahrain - NREN Section
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

3.4 Are there national or institutional open science infrastructures[ix] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

  For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Several initiatives are already in place. The Higher Education Council has highlighted the National Library and Isa Cultural Centre, which contribute to the country's open science efforts. Additionally, Bahrain's open data portal offers a wide range of datasets across various fields such as health, education, and foreign investment, supporting transparency and collaboration in research. This data is accessible to researchers for studies and analysis, furthering Bahrain's commitment to open science.

At the institutional level, universities and think tanks have their own dedicated libraries, journals, and research departments that focus on the exchange of research and dissemination of findings. Furthermore, Derasat notes that there are two national portals—the Bahrain Open Data Portal and the Geographic Information Systems (GIS) National Portal—accessible to both commercial and non-commercial entities. However, measures regarding the CARE and FAIR principles are not explicitly stated.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) has also contributed to Open Science with initiatives such as the "My Digital Library" project, which promotes the publication of digital educational content, and the "Open Resource Portal in Educational Technology and Creativity," which facilitates research collaboration in the fields of educational technology and creativity. These initiatives reflect the growing development of Open Science infrastructures in the Kingdom of Bahrain.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	Yes - The National Library/Isa Cultural Centre: About National Library

	- Bahrain's open data portal provides a wide range of datasets that can be valuable for open science initiatives. The portal includes data on various topics such as health, education, and foreign investment, which can be used by researchers to conduct studies and analyses. By making this data accessible, the Kingdom of Bahrain is contributing to the global movement towards open science and supporting the principles of transparency and collaboration: https://www.data.gov.bh/pages/homepage/ - The infrastructure for open science exists at the level of higher education institutions.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	Under development. At the national level there is the "Bahrain Open Data Portal" and the "Geographic Information Systems (GIS) National Portal". They are accessible for both commercial and non-commercial entities. Measures on CARE and FAIR aspects are not explicity stated and unclear. Links: Bahrain Open Data Portal , GIS National Portal Universities and think tanks have their own dedicated libraries, journals and research department focusing on exchange of research, establishing protocols, disseminating findings, etc.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	- "My Digital Library" Initiative, which is considered one of the digital empowerment projects in education, is concerned with publishing schools' production of digital educational content. This initiative won the ISESCO Award for Open Educational Resources in its first session in 2018 and the Arab Government Excellence Award in the category of "Best Arab Government Project for the Development of Education" in 2023. - RCICT "Open Resource Portal in Educational Technology and Creativity" is a common forum for research institutions with developmental visions and goals. The portal seeks to build research networks in the fields of educational technology and creativity, and to benefit from the expertise of various universities and research institutions. It is also a regional partner platform that allows for the exchange of ideas and information regarding trends, developments, knowledge and expertise related to information and communication technology.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? *(ref.: (iii) 18.e, 18.f)*

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science[i], stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / NO

The Kingdom of Bahrain is involved in regional and international information technology infrastructure for Open Science through various initiatives and collaborations. It hosts the Arabian Gulf University, which operates under the General Secretariat of the Gulf Cooperation Council. This university includes several research centers, such as the Princess Al Jawhara Al Ibrahim Center for Molecular Medicine and Inherited Disorders, where the Arab Association for Genetics Research is headquartered. This association comprises universities, research centers, and scientific institutions from across Arab countries, particularly in the field of genetics.

Bahrain Centre for Strategic, International and Energy Studies (Derasat) highlights the Kingdom's participation in the Arab States Research and Education Network (ASREN), which fosters collaboration among GCC nations in areas of research and information technology. Through these efforts, the Kingdom of Bahrain contributes to the development of federated regional information technology infrastructure for Open Science.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	The Kingdom of Bahrain hosts the Arabian Gulf University, which operates under the General Secretariat of the Gulf Cooperation Council. The university includes a number of research centers, such as Princess Al Jawhara Al Ibrahim Center for Molecular Medicine and Inherited Disorders, where the Arab Association for Genetics Research is headquartered. This association's membership comprises universities, research centers, and scientific institutions from Arab countries active in the field of genetics
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES. It is part of the Arab States Research and Education Network (ASREN), and has established the Regional Centre for ICT, fostering collaboration amongst the GCC nations in areas of research. Links: ASREN

Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? *(ref.: <u>(iii) 18.g</u>)*

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science[i], stage of research, accessibility to all the relevant users in the region or internationally).

The Kingdom of Bahrain is not currently involved in any North-South, North-South-South, or South-South collaborations to optimize infrastructure or develop joint strategies for shared Open Science platforms.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	No
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? *(ref.: <u>(iv) 19.a, 19.c</u>)*

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In the context of systematic capacity building on Open Science within higher education and research institutions in the Kingdom of Bahrain, several initiatives have taken place, primarily focusing on technology access and specialized training programs. The Nasser Centre for Science and Technology (NCST) has been a key player, providing students with free access to technology resources like laptops and iPads to integrate e-learning fully. This investment aims to develop students' technological skills, ensuring they are aligned with the demands of the current and future technological landscape, including the fourth industrial revolution.

Furthermore, specialized training workshops have been a significant part of the capacity-building efforts. These workshops cover advanced topics such as cybersecurity, artificial intelligence, cloud computing, and electrical engineering. The objective is to equip teachers and students with the knowledge and skills to effectively use and apply technology, thereby fostering a culture of open science.

These initiatives also encourage the exchange of ideas and experiences, helping build a community capable of responding to contemporary scientific challenges. An example of this is the educational workshop on enhancing teaching methods with artificial intelligence, conducted in collaboration with the US Embassy.

On the other hand, the Higher Education Council has yet to prioritize Open Science explicitly, focusing more on the quality of the educational services offered. However, they acknowledge that capacity building in open science may be considered in the future.

Similarly, the Bahrain Centre for Strategic, International, and Energy Studies (Derasat) has not explicitly engaged in open science initiatives. There is no defined framework for capacity building in this area, and the institution lacks a vibrant, well-funded research culture to support such efforts.

The Information & eGovernment Authority (iGA) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) have also been involved in related initiatives. RCICT, in particular, has played an active role in hosting workshops and contributing to the development of digital educational content standards and guides in collaboration with the Ministry of Education (MoE).

Notable events include a regional workshop on "Open Educational Resources Policies" held in 2014 and the contribution to the UNESCO book on Open Educational Resources (OER) in 2016, where Bahrain’s experience with OER was highlighted. Additionally, training sessions for teachers and students on digital content production have been conducted, enabling them to create and license educational materials under Creative Commons, further promoting the adoption of Open Science principles.

In summary, while systematic capacity building on Open Science is still evolving in the Kingdom of Bahrain, various institutions, particularly the Nasser Centre and RCICT, have made significant strides in enhancing technological access and fostering Open Science practices through specialized training and workshops. However, broader initiatives are still needed across other institutions to ensure a comprehensive framework for Open Science capacity building in the future.

Nasser Centre for Science and Technology (NCST)	- Access to Technology Resources While reading material and books are available at all times, the focus is on investing in human resources by providing our students free access to technology through offering laptops and/or IPads to integrate e learning fully. This develops their skills and use of technology to align with the current and future technological demands which also encompasses the fourth industrial revolution. - Specialized Training Workshops Specialized training for teachers and students is a key element in achieving the goals of open science. The educational programmes include advanced training in various modern fields such as cybersecurity, artificial intelligence, cloud computing, and electrical engineering. T This training helps equip participants with the skills and knowledge necessary to effectively apply technology, enabling them to transfer this knowledge to their communities and promote sustainable innovation. It also contributes to spreading the culture of open science by encouraging the exchange of ideas and experiences among learners, helping to build a scientifically advanced community that keeps pace with contemporary challenges. An educational workshop on teaching English and the use of artificial intelligence (AI) technologies to enhance teaching methods for the academic staff at the center, in collaboration with the US Embassy.
Higher Education Council (HEC)	No This is because higher education institutions prioritize the quality of educational services they offer to their students. However, capacity building in open science will be considered in the future.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. Not explicity on open science. There is no defined framework for capacity building or formal necessitation towards fostering open science principles. As for this, a key priori is a vibrant and well funded research culture and initiatives, which is lacking.

Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	- A regional workshop entitled: "Open Educational Resources Policies" was held in cooperation with UNESCO, 22-24 September 2014, under the patronage of His Excellency the Minister of Education and chaired by Dr. Miao Fengchen - Chief of the Technology and Artificial Intelligence Unit in Education - UNESCO, Paris. - A contribution to the second chapter of the UNESCO book OER: Policy costs and transformation in 2016 on Kingdom of Bahrain experience with OER. - In collaboration with the Ministry of Education, guides for digital educational content standards were issued. - In collaboration with the Ministry of Education, teachers and students were trained on digital educational content production programmes, resulting in teachers capable of producing digital educational content and licensing it with Creative Commons based on specific criteria. - Introductory sessions were held for different entities in public and private institutions about the Concept of Open Science.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Open Science policies and strategies have not yet been implemented. Nasser Centre for Science and Technology (NCST) has not undertaken such initiatives, and the Higher Education Council has indicated that while they have not yet addressed this area, it will be considered in the future. Similarly, the Bahrain Centre for Strategic, International, and Energy Studies (Derasat) has not focused on integrating Open Science principles into STI policies. The STI policies under Bahrain's Economic Vision 2030 are primarily aimed at enhancing economic growth and the digitization of the economy, with less emphasis on research and open access issues.

The Information & eGovernment Authority (iGA) has not yet been involved in any capacity-building programs for policymakers related to Open Science. However, the UNESCO Regional Centre for Information and Communication Technology (RCICT) has expressed plans to initiate such efforts in the future.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No This will be considered in the future.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. The STI policies under the Bahrain Economic Vision 2030, are tuned towards enhancing economic growth and digitization of the economy and emphasize less on the research and open access aspects.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Planning to be done in the future.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? *(ref.: (iv) 19.b)*

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The incorporation of Open Science competencies into higher education research skills curricula has been addressed to varying degrees across different institutions. At the Nasser Centre for Science and Technology, online counseling platforms help inform students about fields that integrate open science skills into their education.

These platforms provide guidance on how these competencies align with future job opportunities, allowing students to make informed career decisions. Additionally, the center supports this initiative through career counseling workshops and open-door meeting policies.

The Higher Education Council has made progress in integrating research skills into the curricula of academic programmes offered by higher education institutions. This includes promoting an intellectual property culture through academic events for both students and faculty members. Moreover, higher education institutions use software tools such as Turnitin to support the development of research skills, which indirectly supports open science practices.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) has not yet incorporated open science competencies into the higher education research curriculum. The focus at Derasat has primarily been on fostering a strong research culture, with open science initiatives considered as a subsequent policy action.

Similarly, the Information & eGovernment Authority (iGA) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) have not yet implemented or foreseen such frameworks in their initiatives.

In summary, while some institutions like the Higher Education Council and the Nasser Centre have made steps towards integrating Open Science competencies, others have yet to develop or incorporate specific frameworks into curricula, with plans for such initiatives expected in the future.

Nasser Centre for Science and Technology (NCST)	Online counseling platforms can inform students about fields that integrate open science skills within their education. By understanding how these competencies align with future job opportunities, students are better equipped to make strategic career decisions which occurs also through the support provided by the career councelors at the centre through workshops and open door meeting policies.
Higher Education Council (HEC)	Yes Research skills are integrated into the curricula of academic programs offered by higher education institutions. Additionally, intellectual property culture is promoted through organizing academic events for students and faculty members at these institutions. Furthermore, higher education institutions utilize software related to this field such as Turnitin.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. The Higher Education Council puts the emphasis on the research culture itself, and open science initiatives are a subsequent policy action.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

4.4 Are open educational resources[x] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

YES/ PARTLY/ NO

In the Kingdom of Bahrain, Open Educational Resources (OERs) are being used to varying extents as tools for Open Science capacity building. The Nasser Centre for Science and Technology utilizes OERs in online career counseling platforms, offering free and accessible learning materials to promote open science literacy and support informed decision-making for students and parents.

The Higher Education Council has ensured that educational resources, including OERs, are accessible to both academic staff and students for educational and scientific research purposes. The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) partly utilizes OERs, with the primary focus on open data provision. Additionally, the UNESCO Regional Centre for Information and Communication Technology (RCICT) actively supports the use of OERs to advance Open Science initiatives.

Nasser Centre for Science and Technology (NCST)	Online career counseling can utilize Open Educational Resources (OERs) to provide free and accessible learning materials for students and parents. By integrating OERs, the platform promotes open science literacy, making valuable content available to a wider audience and supporting informed decision-making for students' educational and career paths.
Higher Education Council (HEC)	Yes At higher education institutions, all educational resources are accessible to their members, including academic staff and students, for educational or scientific research purposes.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	PARTLY. The only relevant action in this regard is the open data provision.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Yes

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? *(ref.: (iv) 19.e)*

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Several initiatives have been launched to support science communication in line with Open Science values, aiming to disseminate scientific knowledge to researchers, decision-makers, and the public. At the Nasser Centre for Science and Technology (NCST), online career counseling initiatives provide resources that explain the role of open science in various careers, helping students and parents understand how open science contributes to different industries and guides career choices. The Higher Education Council (HEC) has implemented multiple initiatives to promote scientific communication, including fostering a culture of scientific research through national competitions and supporting research capabilities in both schools and higher education institutions. The Council also organizes numerous peer-reviewed scientific conferences that bring together researchers, decision-makers, business leaders, and other stakeholders. Furthermore, efforts are underway to establish a national research network to better connect researchers, decision-makers, and other interested parties.

The Bahrain Centre for Strategic, International, and Energy Studies (Derasat) also supports science communication by facilitating the dissemination of scientific knowledge, though specific initiatives were not detailed.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) has created the "Open Resource Portal in Educational Technology and Creativity," which serves as a platform for research institutions to exchange knowledge and build research networks.

This portal connects various universities and institutions to share trends, developments, and expertise in the fields of educational technology and creativity, further supporting science communication in line with Open Science principles.

Nasser Centre for Science and Technology (NCST)	Online career counseling can benefit from science communication initiatives by providing parents and students with access to resources that explain open science's role in careers. This helps both students and parents understand how open science contributes to various industries and informs their career path choices.
Higher Education Council (HEC)	Yes. Several initiatives have been implemented to support scientific communication among various groups, including: - Promoting a culture of scientific research among school students and higher education institutions, organizing national competitions, and supporting research capabilities. - Organizing numerous peer-reviewed scientific conferences by higher education institutions in various fields, inviting researchers, decision-

	makers, business leaders, and other stakeholders interested in the topics of these conferences. Additionally, efforts are underway to establish a national research network aimed at connecting researchers, decision-makers, and interested parties.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	Yes
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	RCICT "Open Resource Portal in Educational Technology and Creativity" is a common forum for research institutions with developmental visions and goals. The portal seeks to build research networks in the fields of educational technology and creativity, and to benefit from the expertise of various universities and research institutions. It is also a regional partner platform that allows for the exchange of ideas and information regarding trends, developments, knowledge and expertise related to information and communication technology.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: <u>(v) 20.b, 20c, 20i</u>)*

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Several initiatives have been undertaken at the national and institutional levels to review and potentially align research assessment and evaluation processes with Open Science principles, or such initiatives are foreseen by 2025.

The Higher Education Council has made progress in this area by ensuring that higher education institutions prepare annual research plans that include priority areas, many of which are related to Open Science.

These institutions have dedicated units or deanships responsible for managing and evaluating research affairs, with evaluation criteria that incorporate scientific research principles. Furthermore, the Council also reviews the internal regulations of higher education institutions, including research policies and procedures, to ensure they support the principles of open science.

However, the Bahrain Centre for Strategic, International and Energy Studies (Derasat) has not yet taken steps to align research assessment processes with open science values. The focus remains on advancing research output and building infrastructure to foster a culture of innovation, with the integration of open science into research assessment being a forthcoming step.

Additionally, the UNESCO Regional Centre for Information and Communication Technology (RCICT) is working on a training programme for K-12 teachers on Open Science, which will provide credit hours upon completion, helping teachers in their career advancement and indirectly supporting the broader integration of Open Science principles in education and research.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	Yes Higher education institutions prepare annual research plans that include priority research areas, most of which are related to open science. These institutions also include a deanship or unit within their organizational structure responsible for managing and organizing research affairs within the institution. One of its roles is to evaluate the research projects submitted by faculty members, with evaluation criteria that include scientific research principles. In addition, the higher education institutions have their procedures to review the research plans submitted by graduate students. The General Secretariat of the Higher Education Council also reviews the internal regulations of higher education institutions, including research policies, regulations, and procedures, considering that they support the principles of open science.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. National policies and institutional priorities, are still placed towards advancing the research output and building the required infrastructure to foster a culture of innovation and research. The integration of open science into research assessment processes, is the forthcoming step towards solidifying the academic and research culture in the nation.

Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	Working on preparing a training programme about Open Science (OS) for K12 teachers and they will receive credit hours upon completion of the training which helps them in their career advancement.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? *(ref.: (v) 20.a, 20.c, 20.d)*

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In the Kingdom of Bahrain, the incorporation of clauses that recognize Open Science practices in career evaluation and progression systems is still in development, with some institutions considering it for future action. The Higher Education Council has not yet integrated specific clauses recognizing open science practices, but discussions are underway for future collaboration with higher education institutions.

At the Bahrain Centre for Strategic, International, and Energy Studies (Derasat), some steps have been taken at the institutional level to foster cross-country and cross-cultural collaborations. The National Research Strategy 2014-2024 includes provisions for the efficient and fair dissemination of research outputs to relevant stakeholders, though no stringent clauses or frameworks are currently in place regarding open science awareness, development of assessment systems, or cross-sectional stakeholder engagement.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) has created the "Open Resource Portal in Educational Technology and Creativity," which serves as a forum for research institutions and facilitates collaboration.

This platform supports research networks in educational technology and creativity, enabling the exchange of ideas, trends, developments, and expertise, thus indirectly encouraging collaboration with societal actors beyond the scientific community. However, formal integration of Open Science practices into career evaluation and progression systems is still a work in progress, with plans for further development expected in the future.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No This is being considered for future action in collaboration with higher education institutions.

Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES. At the institutional level, entities have established cross-country and cross-cultural collabrations. Provisions on efficient, fair and rigourous dissemination of research output to all the relevant stakeholders have been put forward partially in the Natioanal Research Strategy 2014-2024. However, no stringent clauses or frameworks are in place for the foreseeable future, towards the awareness of open science, development of assessment systems and cross-sectional stakeholder engagement.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	RCICT "Open Resource Portal in Educational Technology and Creativity" is a common forum for research institutions with developmental visions and goals. The portal seeks to build research networks in the fields of educational technology and creativity, and to benefit from the expertise of various universities and research institutions. It is also a regional partner platform that allows for the exchange of ideas and information regarding trends, developments, knowledge and expertise related to information and communication technology.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? *(ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)*

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders[xi] that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In the Kingdom of Bahrain, there have been some initiatives to recognize and reward research practices that align with open science principles, though these are still in the early stages. The Higher Education Council has implemented several initiatives based on existing legislative texts, which encourage research publication.

Some higher education institutions offer incentives to faculty members who publish scientific research, with the level of incentive varying depending on the prestige of the publication. Additionally, these institutions organize annual events to recognize outstanding research and honour the researchers behind them. While these efforts are ongoing, they are gradually being considered by other higher education institutions, suggesting that Open Science recognition may become more widespread in the future.

However, at the Bahrain Centre for Strategic, International, and Energy Studies (Derasat), open science is still in the prescriptive phase, and initiatives to formally incorporate it into the national research strategy are yet to be established. Similarly, the Information & eGovernment Authority (iGA) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) have not yet introduced specific initiatives in this area.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	Yes Several initiatives have been implemented based on legislative texts rather than the concept of open science. In addition, some higher education institutions are offering incentives to faculty members who publish their scientific research, with the incentives varying according to the level of the publication. Additionally, these institutions organize annual events to recognize outstanding research and the researchers behind them. While this is happening, it is also being actively considered by other higher education institutions.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. The talks on open science are yet at the prescriptive stage, and are yet to be incorporated in the national research strategy and aims, which follows after a thriving research culture and environment.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

5.4 Have there been any unintended negative consequences [xii] of open science practices reported in your country? (ref.: (v) 20)

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

No unintended negative consequences of open science practices have been reported. The Higher Education Council has stated that no negative actions related to Open Science practices have occurred. Similarly, the Bahrain Centre for Strategic, International and Energy Studies (Derasat) has not observed any negative consequences.

The Information & eGovernment Authority (iGA) and the UNESCO Regional Centre for Information and Communication Technology (RCICT) have not reported any such issues either. Overall, Open Science practices have not led to any unintended negative outcomes in the country so far.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	No No negative actions related to open science practices have been reported.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	No
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	No

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.

- promoting open innovation practices.
- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Several initiatives at both the national and institutional levels aim to promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the scientific process. The Higher Education Council has led efforts to explore open peer-review practices, develop participatory strategies for identifying the needs of marginalized communities, and promote open innovation practices. Additionally, higher education institutions have been directed to revise undergraduate study plans to include courses that develop scientific research skills. For postgraduate students, institutions provide detailed guides to assist with the entire thesis process, from research planning to thesis completion. These guides are published on institutional websites to ensure accessibility. Students also engage in seminar sessions where they receive feedback from faculty members on their research plans, fostering a collaborative approach.

After completing their thesis, postgraduate students present and defend their research in public sessions, incorporating feedback before submitting final copies to university libraries and the National Library, making them accessible to students, researchers, and the public. The Council also encourages a focus on research in priority areas and provides software tools like SPSS for data analysis and Turnitin for plagiarism detection.

At the Bahrain Centre for Strategic, International, and Energy Studies (Derasat), cross-sector collaboration between academia and industry is encouraged to develop a scientific culture and innovative research processes. While there are no formal policies mandating Open Science practices, the research bodies at institutions promote such approaches indirectly. The UNESCO Regional Centre for Information and Communication Technology (RCICT) also supports collaboration among different departments within the Ministry of Education, as well as with public and private educational and research institutions, to foster a scientific culture and enhance research practices.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	YES, at both national and institutional levels. - exploration of open peer-review practices. - development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas. - promoting open innovation practices. - Others • Higher education institutions have been directed to review the study plans for undergraduate programmes to include a course that aims to develop scientific research skills. • Higher education institutions, particularly those offering postgraduate programmes, have developed guides to assist students

	in preparing their theses. These guides cover the entire process, from drafting the research plan to completing the thesis, and are typically published on the institutions' websites. • For postgraduate students, the process begins with the submission of a research plan that outlines the required components of the thesis. This plan is reviewed during seminar sessions organized by academic departments, where students are encouraged to incorporate feedback from faculty members and discuss the plan's content with them. • Once a student completes his thesis, a public defense session is held where he presents and discusses the methodology and content of his work. After incorporating any required revisions, printed copies of the thesis are deposited in the university library to be available for review by library users. They may also be reproduced in accordance with library regulations and intellectual property rights. A copy of the thesis shall also submitted to the National Library (public library), which makes it accessible to students, researchers, and the public. • Academic staff members and students are encouraged to focus on research in priority areas. • Higher education institutions provide software tools to support researchers, such as the SPSS programme for data analysis and Turnitin for detecting academic plagiarism.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	YES. At the institutional level, cross-sector collaboration between academia and industry have been in place to develop a scientific culture and research processes. Although there are no mandated policies for the purpose, the bodies overseeing research at institutions, ensure and promote innovative approaches for open science, indirectly.

Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	YES. At the institutional level, collaboration between different departments at the Ministry of Education, public and private educational and research inststutions have been in place to develop a scientific culture.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? *(ref.: (vii) 22.a)*

The Kingdom of Bahrain promotes and facilitates international scientific collaborations on Open Science through various initiatives at both the institutional and national levels. The Higher Education Council encourages higher education institutions to establish memoranda of understanding (MOUs) and cooperation agreements with renowned international institutions. These agreements focus on joint research projects, exchanging research findings, providing access to databases, and promoting academic mobility for both faculty members and students.

At the Bahrain Centre for Strategic, International, and Energy Studies (Derasat), international collaboration ties are fostered through partnerships with global research bodies, facilitating scientific and cultural exchanges. Additionally, foreign universities are involved in establishing state-of-the-art research facilities and cultivating a robust research culture in the Kingdom of Bahrain.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) is also actively involved in promoting international scientific collaborations. It partners with the International Research Centre on Artificial Intelligence (IRCAI) in Slovenia, affiliated with UNESCO, and participates in the Open Education for a Better World (Oe4bw) programme, which develops Open Educational Resources (OER) to support the dissemination of Open Science concepts at local, regional, and international levels.

RCICT also participated in a regional conference in November 2024 titled "Democratizing Science in the Arab Region," organized by the UNESCO Multisectoral Regional Office in Rabat, Morocco, which highlighted the importance of building infrastructures for open access to educational and informational resources.

Furthermore, RCICT contributed to the "Digital Learning Resources Forum Opportunities and Challenges" by presenting two papers on Open Science (OS) and Open Educational Resources (OER) to an audience of 70 beneficiaries from GCC countries and Yemen.

Nasser Centre for Science and Technology (NCST)	N/A
Higher Education Council (HEC)	In line with the cooperation agreements and memorandom of understanding signed between the Kingdom of Bahrain and other countries, the Higher Education Council encourages higher education institutions to enter memoranda of understanding and

	cooperation agreements with renowned international institutions. These agreements should focus on areas such as conducting joint research, exchanging research findings, providing access to databases, and promoting academic mobility for faculty members and students.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	By collobrating with international research bodies at the institutional and national levels, scientific and cultural exchange are developed, in addition to fostering a culture of research and academia.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	- In light of RCICT's partnership with the International Research Centre on Artificial Intelligence (IRCAI) in Slovenia, affiliated with UNESCO, the Centre regularly participates in the Open Education for a Better World (Oe4bw) programme, which focuses on developing OER that support identifying and disseminating the concept at local, regional and international levels. - RCICT participated in a regional conference on November 28 and 29, 2024, titled " Democratizing Science in the Arab Region", which showcased the the key developments and practices in support of Open Science (OS) in the Arab region, and focused on the importance of building infrastructures that support open access to educational and information resources. - This event was organised by the UNESCO Multisectoral Regional Office in Rabat, Morocco, in cooperation with the Arab League Educational, Cultural and Scientific Organization (ALECSO), UNESCO and Mohammed V University in Rabat. - RCICT participated in "Digital Learning Resources Forum Opportunities and Challenges "organized by Arab Bureau of Education, RCICT and the Ministry of Education for 70 beneficiaries from GCC countries and Yemen by presenting two papers in the

	discussion sessions about Open Science (OS) and Open Educational Resources (OER).

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? *(ref.: (vii) 22.b, 22.e, 22.f)*

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The Kingdom of Bahrain is actively fostering cross-border multi-stakeholder collaboration on open science through several strategic initiatives. The Higher Education Council has outlined a National Strategy for Scientific Research (Bahrain-Research-Strategy-2014-2024) that focuses on enhancing integration with international academic and research institutions. This strategy promotes partnerships with global entities to address Bahrain's economic and social priorities and includes initiatives aimed at raising public awareness about research and innovation. One key aspect of this strategy is monitoring scientific research indicators, such as joint research initiatives with regional and international universities.

The Council has facilitated numerous collaborations, including organizing a Research Boot Camp in partnership with the British Council to equip researchers with essential skills in areas like research dissemination, digital tools for analysis, and project management. It also encourages universities to establish memoranda of understanding (MOUs) with international institutions, and it organizes forums and field visits to promote international collaboration in scientific research.

In addition, the Nasser Centre for Science and Technology (NCST) supports cross-border collaborations by engaging in regional networks and partnerships that focus on vocational education and training. These partnerships involve multi-stakeholder collaboration, requiring expertise in open science tools and methods to facilitate knowledge exchange and joint capacity-building efforts.

The UNESCO Regional Centre for Information and Communication Technology (RCICT) also plays a significant role by supporting UNESCO's initiatives and delivering free regional training programmes on Open Science (OS) in collaboration with the Ministry of Education. This program is designed to disseminate the culture of Open Science across the region. However, the Bahrain Centre for Strategic, International and Energy Studies (Derasat) has not yet implemented specific policies or plans for cross-country collaboration on Open Science.

Nasser Centre for Science and Technology (NCST)	IPD is actively fostering cross-border collaborations by engaging in regional networks and initiatives that support vocational education and training. This includes partnerships with universities and the development of multi-stakeholder partnerships both locally and internationally. These collaborations require technical expertise in open science tools and methods, facilitating knowledge exchange and joint capacity-building efforts.

Higher Education Council (HEC)	The National Strategy for Scientific Research ((Bahrain-Research-Strategy-2014-2024_Arabic.pdf) approved by the Higher Education Council, outlines several goals, including the third strategic objective: enhancing integration between international academic and research institutions and entities that focus on Bahrain's economic and social priorities, as well as raising public awareness about research and innovation. The Higher Education Council also monitors scientific research indicators, including research projects funded by regional and international entities, and joint research initiatives with regional and international universities or research centers. Throughout the Council's operational plans, various initiatives and projects have been carried out in collaboration with international organizations, such as the American Association for the Advancement of Science (AAAS), the British Embassy, and the British Council. Examples of these initiatives are: - Organizing Research Boot Camp in collaboration with the British Council. The Boot Camp served as a platform to equip both early and advanced researchers with essential skills and knowledge. The participants were engaged in enriching sessions focused on key areas such as research dissemination, the use of digital tools for analysis, networking and collaboration, project management, and commercialization of research - Expanding collaboration opportunities between higher education institutions in the Kingdom of Bahrain and global institutions by encouraging universities to sign memoranda of understanding and cooperation agreements in the field of scientific research, host academic programmes, and open branches of foreign higher education institutions in the Kingdom of Bahrain. Additionally, forums are being held with representatives from Bahraini and foreign higher education institutions, such as a forum co-organised with Canadian higher education institutions and a meeting jointly held with the University of Cyprus. - Organizing a field visit for deans and officials responsible for scientific research at government, regional, and private higher education institutions to research institutions and universities in the United Kingdom. The visit was held in collaboration with the Science and Innovation Network in the UK, the British Council, and the British Embassy to the Kingdom of Bahrain.

Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. There are no explicit policy actions or plans in place towards cross-country collaboration on open science. The main hurdles, being that this has not held the required priority as compared to other broader national aims.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	The RCICT contributed to this drive aimed disseminting the culture of Open Science (OS) by delivering a free of charge regional training programme about (OS) after getting the scheme accredited by competent entities in the Ministry of Education.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: *(vii) 22.c)*

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, the Kingdom of Bahrain is not directly involved in discussions regarding the establishment of regional and international funding mechanisms for Open Science. The Nasser Centre for Science and Technology (NCST), while not actively engaged in such discussions, stands to benefit from existing and future initiatives that align with Open Science objectives.

The NCST is particularly focused on skills development and technical training to enhance Bahrainis' participation in international collaborations, positioning itself as a viable hub and key partner in future open science frameworks.

Similarly, the Higher Education Council has not been involved in these discussions, although there have been consultations with external parties to explore cooperation in the areas of education and scientific research. The Bahrain Centre for Strategic, International and Energy Studies (Derasat) also reports no involvement in these discussions.

Nasser Centre for Science and Technology (NCST)	NO. While IPD may not be directly engaged in certain discussions, it stands to benefit from existing and future initiatives aligned with open science objectives. IPD is actively focused on skills development and technical training to enhance Bahrainis' participation in international collaborations, positioning NCST as a key partner in establishing the necessary frameworks.

Higher Education Council (HEC)	No There are consultations with some external parties to open areas of cooperation related to education and scientific research.
Bahrain Centre for Strategic, International and Energy Studies (Derasat)	NO. This can again be attributed to the nascent research and scientific culture in the region as a whole. Furthermore, the research budget has been below the global average, suggesting impediments towards any funding mechanisms.
Information & eGovernment Authority (iGA)	--
UNESCO Regional Centre for Information and Communication Technology (RCICT)	--

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO

i Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

ii The 2021Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

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